and the Big Bang hypothesis for the formation of the universe as originally suggested in 1948 by Gamow, Alpher and Herman, and later developed by the B^2FH-group, Weinberg and others.

12.2. IN THE BEGINNING OF TIME

Around "time zero" the Universe consisted of an immensely dense, hot sphere of photons, quarks and leptons, and their antiparticles, in thermal equilibrium, particles being created by photons and photons created by annihilation of particles. The temperature must have been $\geq 10^{13}$ K, but no light was emitted, because of the enormous gravitational force pulled the photons back. The system was supposed to be in a unique state with no repulsion forces. However, just as a bottle of supercritical (overheated) water can explode by a phase transition, so did the Universe, space came into being and time began. The Universe expanded violently in all directions, and as age and size grew, density and temperature fell.

A one hundreds of a second later all the quarks were gone, and the Universe consisted of an approximately equal number of electrons, positrons, neutrinos and photons, and a small amount of protons and neutrons; the ratio of protons to photons is assumed to have been about 10^{-9}. The temperature was about 10^{11} K and the density so high, about $4 \times 10^6\,\mathrm{kg\,m^{-3}}$, that even the unreactive neutrinos were hindered to escape.

The conditions can be partly understood by considering the relations

$$E\,(\mathrm{MeV}) = m\,\mathbf{c}^2 = 931.5\;\Delta M\,(\mathrm{u}) \qquad (4.6 \text{ and } 12.8)$$

which gives the energy required to create a particle of mass ΔM, and

$$E\,(\mathrm{MeV}) = \mathbf{k}\;T = 8.61 \times 10^{-11}\,T \qquad (3.53 \text{ and } 12.9)$$

which gives the average kinetic energy of a particle at temperature T (K, Kelvin). As the photon energy of E (eV) corresponds to the wavelength λ (m) according to (3.52)

$$E\,(\mathrm{eV}) = \boldsymbol{h}\,\nu = \boldsymbol{h}\,\mathbf{c}/\lambda = 1.240 \times 10^{-6}/\lambda\,\left(\mathrm{m}\right) \qquad (12.10)$$

one can estimate that the creation of a proton or a neutron (rest mass 940 MeV) out of radiation requires a temperature of 1.1×10^{13} K, corresponding to a photon wave length of about 10^{-15} m, i.e. the size of a nucleon. At these temperatures nucleons are formed out of radiation, but are also disrupted by photons, leading to an equilibrium with about an equal number of protons and neutrons. At temperatures below the threshold formation energy, no nucleons are formed. However, it should be remembered that particles and radiation are distributed over a range of energies, according to the Boltzmann (cf. Fig. 3.4) and Planck distribution laws. Thus some formation (and disruption) of nucleons occurs even at lower temperatures.

In about 0.1 s the temperature is assumed to have decreased to 3×10^{10} K (corresponding to 2.6 MeV). Now, the equilibrium between protons, neutrons, electrons and neutrinos can be written

$$p + \nu \rightleftharpoons n + e^{+} \quad Q = -1.80 \text{ MeV} \tag{12.11}$$

$$n + \nu \rightleftharpoons p + e^{-} \quad Q = 0.78 \text{ MeV} \tag{12.12}$$

The mass of the neutron exceeds that of the proton by a small margin of 0.0013884 u, corresponding to 1.293 MeV; thus reaction (12.11) requires energy, while reaction (12.12) releases energy. The formation of protons was therefore favoured over neutrons, leading to 38% neutrons and 62% protons.

As temperature and density further decreased the neutrinos began to behave like free particles, and below 10^{10} K they ceased to play any active role in the formation sequence (matter became transparent to the neutrinos). The temperature corresponded then to ~1 MeV, i.e. about the threshold energy for formation of positron/electron pairs. Consequently they began to annihilate each other, leaving, for some reason, a small excess of electrons. Though neutrons and protons may react at this temperature, the thermal energies were still high enough to destroy any heavier nuclides eventually formed.

After 14 s the temperature had decreased to 3×10^{9} K (0.27 MeV) and 3 min later to about 10^{9} K (<0.1 MeV). Now, with the number of electrons, protons and neutrons about equal (though the universe mostly consisted of photons and neutrinos), some protons and neutrons reacted to form stable nuclides like deuterium and helium:

$$^{1}H + {}^{1}H \rightarrow {}^{2}H + \left|e^{-} + \beta^{+}\right| + \nu_e \quad Q = 0.42 + 1.02 = 1.44 \text{ MeV} \tag{12.13}$$

This reaction is exothermic. However, for the ^{2}H to be stable the temperature must decrease below the Q-value, i.e. to about 10^{10} K, and, in reality, to the much lower value of about 10^{9} K, because of the high photon flux which may dissociate ^{2}H into 2 ^{1}H. Two deuterium atoms then fuse, probably in several steps as discussed below, to form He:

$$^{2}H + {}^{2}H \rightarrow {}^{4}He \quad Q = 23.85 \text{ MeV} \tag{12.14}$$

^{4}He is an extremely stable nucleus, not easily destroyed, as compared to nuclides with masses >4, whose binding energies (per nucleon) are only a few MeV (Fig. 4.3). As the universe expanded, the probability for particle collisions decreased, while the kinetic energy available for the fusion reactions was reduced. Therefore, the nucleon build-up in practice stopped with ^{4}He, leading to an average universal composition of 73% hydrogen and 27% helium. A very small amount of deuterium atoms was still left, as well as a minute fraction of heavier atoms, formed by the effects of the "Boltzmann tail" and "quantum tunnelling" (see §12.4 and Fig. 12.6). The remaining free neutrons (half-life 10.4 min) now decayed to protons.

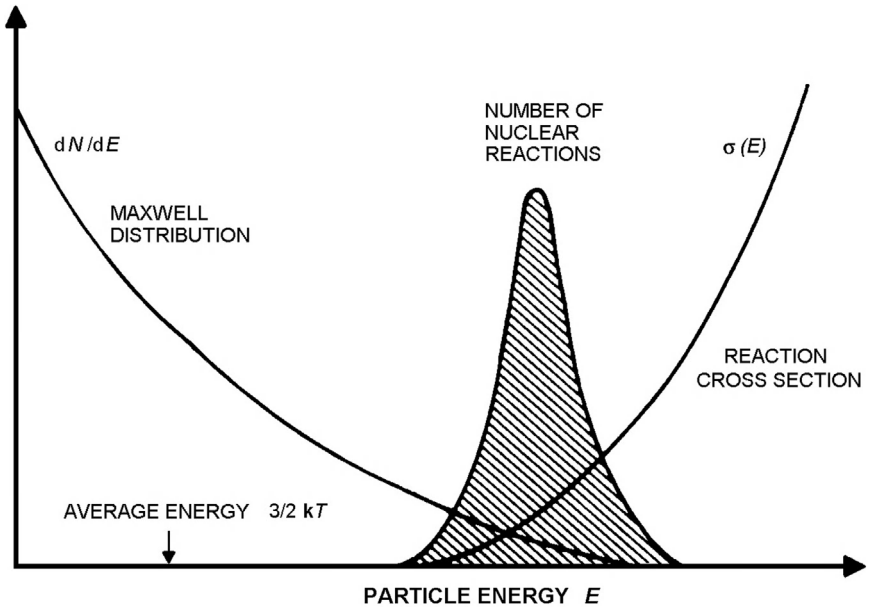


Figure 12.6 Fusion reactions occurring in the shaded area, as a function of particle energy, showing Maxwell particle distribution and reaction cross section.

The situation about 35 minutes after time zero was then the following: Temperature was 3×10^8 K, density about 10^{-4} kg m^{-3}. The Universe consisted of 69% photons, 31% neutrinos, and a fraction of 10^{-9} of particles consisting of 72–78% hydrogen and of 28–22% helium, and an equivalent number of free electrons, all rapidly expanding in all directions of space.

It was still too hot for the electrons to join the hydrogen and helium ions to form neutral atoms. This occurred not until about 500 000 years later, when temperature had dropped to a few 1000 K. The disappearance of free electrons broke the thermal contact between radiation and matter, and radiation continued then to expand freely. An outside spectator would have observed this as a hugh flash and a rapidly expanding fireball. In the adiabatic expansion the radiation cooled further to the cosmic background radiation level of ~2.7 K measured today.

The recent observation that the cosmic background radiation shows ripples in intensity in various directions of space indicates a slightly uneven ejection of matter into space, allowing gravitational forces to act, condensing the denser cloud parts into even more dense regions, or "islands", which by time separated from each other, leaving seemingly empty space in between. Within these clouds, or proto-galaxies, local higher densities lead to the formation of stars, as will be explained in next section.

12.3. STAR IGNITION

The condensation of matter releases gravitational energy which is transformed into kinetic energy of the particles in the gas cloud, increasing temperature and pressure. As the density increases, radiation transport through the cloud becomes more difficult, leading to more rapid heating of the gas.

At a critical density, the *Chandrasekhar limit*, the hydrogen degenerates into H^+ ions and non-localized ("free") electrons (this is the *plasma state*). The ionization energy of H (13.6 eV) corresponds to an average kinetic particle energy of 160 000 K, eqn. (3.31), but because of the particle energy distribution (Fig. 3.4), degeneration begins already at lower temperature. Because the elimination of the electron shell leads to a pressure reduction, the gravitational forces cause an increased rate of condensation, with a further rise in temperature until eventually all the gas is ionized.

Because the temperature decreases from the center towards the shell, the light emitted from the center may be absorbed in outer layers. Thus the protostar may be invisible, but as it gradually heats up it begins to emit light. There are many such faint objects in our galaxy.

Depending on the mass of the contracting cloud, the star evolution follows different paths.

(i) For a cloud with $m < 0.08\ \mathbf{M}_\odot$, temperature and pressure never reach high enough values for hydrogen ignition and such "stars" will contract to planet-like *brown dwarfs.*

(ii) For masses $>0.08\ \mathbf{M}_\odot$ fusion reactions begin when the core reaches $\geq 4 \times 10^6$ K. In Figure 12.4 we have indicated the birth of a star like our sun ($1\ \mathbf{M}_\odot$) which appears initially as a red object in the sky, indicating a low surface temperature, but the luminosity slowly increases over a condensation time of $10^5 - 10^6$ y.

The Coulomb barrier for interaction of protons with each other is 1.1 MeV, eqn. (10.15). In the sun, density and temperature increases strongly from the surface towards the core. At a core temperature of 1.4×10^7 K, the average kinetic energy of a proton is only 1.8 keV (the most probable energy is 1.2 keV, § 3.6.2, while the fraction of protons with energies $\geq$1.1 MeV is only $\sim 10^{-398}$, eqn. (3.35). Formation of a ^{2}He nucleus from two protons is endothermic and will not occur by tunnelling due to the lack of a driving force. On the other hand the reaction forming ^{2}H is exothermic. Hence, quantum mechanical tunnelling can allow two protons to interact to form a ^{2}H nucleus according to (12.13) during the brief encounter of a scattering event even at energies $\ll$1.1 MeV. The reaction cross section is extremely small due to the involvement of the weak force and a change in spin for one of the nucleons ($\sim 10^{-47}$ barn at 1 MeV collision energy). Once deuterium has been formed it may fuse into helium. Thus the star has begun its "childhood" by burning hydrogen into helium: it has entered the Zero Age Main Sequence of stars. Stars with smaller masses enter lower on the Main Sequence, and evolve very slowly.